

servlets

A Servlet is a Java class used to build Dynamic web applications. It runs on a server and handles client requests (like from a browser), then returns responses (like an HTML page or JSON).

How It Works?

- > Inside a Servlet Container (like Apache Tomcat)
- > Part of Java EE / Jakarta EE

what can do using servlets?

- > Receive HTTP requests (GET, POST, etc.)
- > Process data (like querying a database)
- > Send back responses (HTML, JSON, text, etc.)
- > Manage sessions, cookies, etc.

Servlet Life Cycle

init() → Runs once when servlet is created

service() → Runs for every request

destroy() → Runs once when servlet is removed

Servlet (interface)

GenericServlet implements Servlet (abstract class)

HttpServlet extends GenericServlet (abstract class)

- > doGet
- > doPost
- > doPut
- > delete

Servlet (interface)

@Override

```
public void destroy() {  
  
}
```

@Override

```
public ServletConfig getServletConfig() {  
    return null;  
}
```

```

@Override
public String getServletInfo() {
    return null;
}

@Override
public void init(ServletConfig config) throws ServletException {

}

@Override
public void service(ServletRequest req, ServletResponse res) throws
ServletException, IOException
{

}

```

ServletConfig -> Interface

ServletConfig gives a servlet access to:

- > get parameters from defined in web.xml
- > The ServletContext
- > The servlet's name

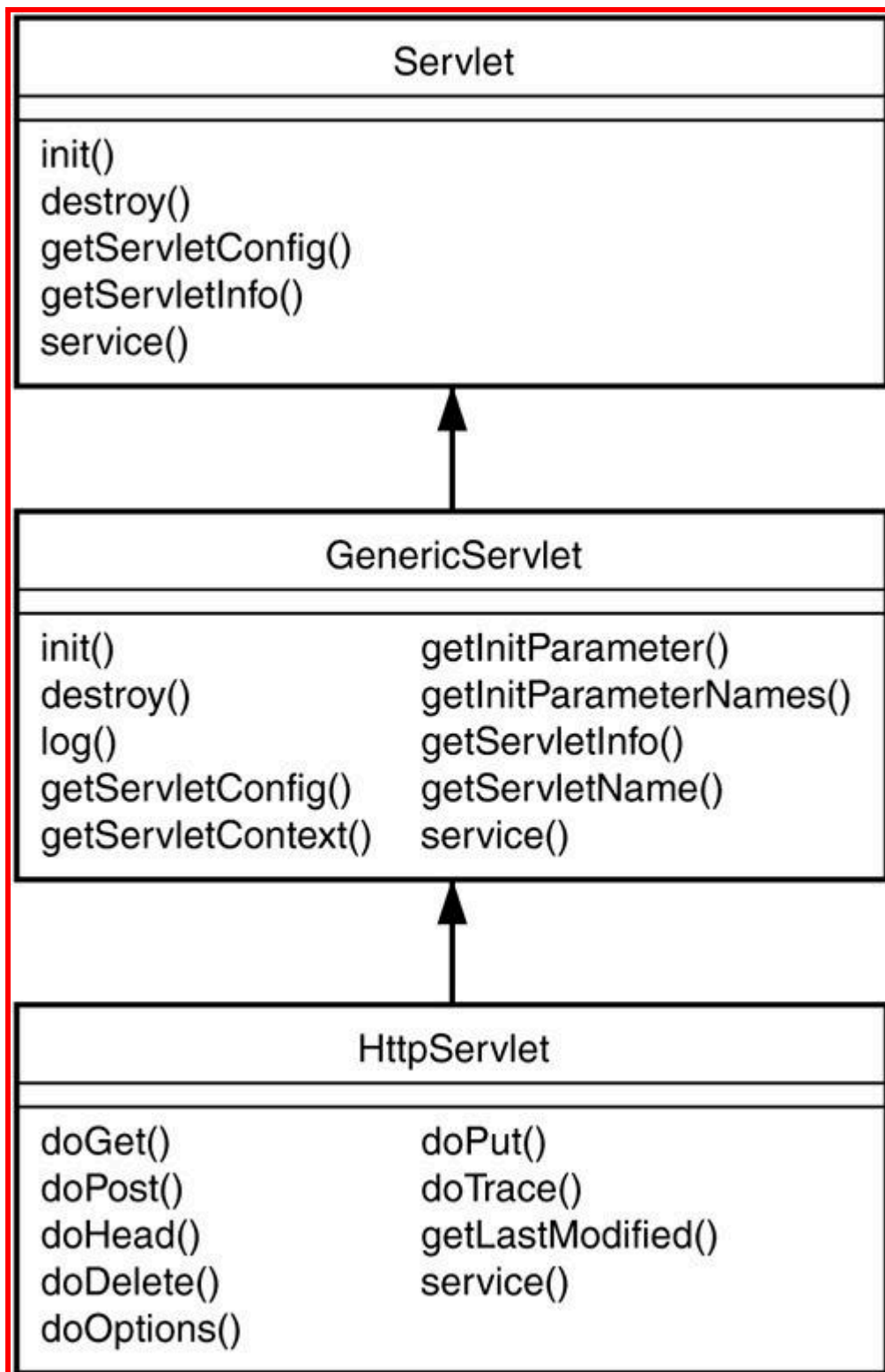
This is useful when you want to pass configuration values (like DB URLs, usernames, or other settings) without hardcoding them in your servlet.

HTTP Methods:

HTTP methods, also known as HTTP request methods, are instructions that a client (like a web browser) sends to a server to perform actions on a resource, such as retrieving, creating, updating, or deleting data.

The most common methods are

- > GET - Retrieves The data
- > POST - Create
- > PUT - update
- > DELETE - delete data



Servlet Life cycle program

```
package EnpoinConnection;

import jakarta.servlet.*;
import jakarta.servlet.annotation.WebServlet;
import java.io.IOException;

@WebServlet("/lifeCycle")

public class ServletLifeCycle implements Servlet {
    ServletConfig config;

    @Override
    public void init(ServletConfig config) throws ServletException {
        System.out.println("run init");
        this.config=config;
    }

    @Override
    public ServletConfig getServletConfig() {
        return config;
    }

    @Override
    public void service(ServletRequest req, ServletResponse res) throws
ServletException, IOException {

        System.out.println("run service");
    }

    @Override
    public String getServletInfo() {
        return "";
    }

    @Override
    public void destroy() {

        System.out.println("run destroy");
    }
}
```

GenericServlet Program

```
package EnpoinConnection;

import jakarta.servlet.GenericServlet;
import jakarta.servlet.ServletException;
import jakarta.servlet.ServletRequest;
import jakarta.servlet.ServletResponse;
import jakarta.servlet.annotation.WebServlet;
import java.io.IOException;

@WebServlet("/generic")
public class GenericServletPractice extends GenericServlet {

    @Override
    public void init(){
        System.out.println("override the init");
    }

    @Override
    public void service(ServletRequest req, ServletResponse res) throws
    ServletException, IOException {
        System.out.println("Handle the all method");
    }
}
```

HttpServlet program

```
package EnpoinConnection;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;

@WebServlet("/HttpServlet")
public class HttpServletPractice extends HttpServlet {
    @Override
    public void doGet(HttpServletRequest req, HttpServletResponse resp)
        throws ServletException, IOException {
        System.out.println("get method");
    }

    @Override
    protected void doPost(HttpServletRequest req, HttpServletResponse
resp) throws ServletException, IOException {
        System.out.println("post method");
    }

    @Override
    protected void doPut(HttpServletRequest req, HttpServletResponse
resp) throws ServletException, IOException {
        System.out.println("put method");
    }

    @Override
    protected void doDelete(HttpServletRequest req, HttpServletResponse
resp) throws ServletException, IOException {
        System.out.println("Delete method ");
    }

    @Override
    protected void service(HttpServletRequest req, HttpServletResponse
resp) throws ServletException, IOException {
        System.out.println("this is service method");
    }
}
```

GetParametr method and BufferReade

```
package EnpoinConnection;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

import java.io.BufferedReader;
import java.io.IOException;
@WebServlet("/get")
public class GetParam extends HttpServlet {
    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse
resp) throws ServletException, IOException {
//        super.doGet(req, resp);

        String username=req.getParameter("username");
        String password=req.getParameter("password");
        System.out.println(username+ " - " + password);
//
http://localhost:8080/ServletEnpoint/get?username=mango&password=banana

        BufferedReader bf=req.getReader();
        int ch;
        while ((ch=bf.read())!=-1){
            System.out.println((char) ch);
        }
        //past this is post man :
http://localhost:8080/ServletEnpoint/get
//        {
//            "id": 1,
//            "name": "resh",
//            "email": "resh@1205",
//            "price": 4567
//        }

        System.out.println("Client IP : " + req.getRemoteAddr());

        System.out.println("Protocol : " + req.getProtocol());

        System.out.println("Server Name : " + req.getServerName());

        System.out.println("Server Port : " + req.getServerPort());

        System.out.println("Request URL : " + req.getRequestURL());

        System.out.println("Request URI : " + req.getRequestURI());
```

```

        System.out.println("Context Path : " + req.getContextPath());

        System.out.println("Method : " + req.getMethod());
    }
}

```

Content Type program

```

package EnpoinConnection;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

import java.io.IOException;
import java.io.PrintWriter;

@WebServlet("/content")
public class ContentType extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse
res) throws ServletException, IOException {

        ///-----text/html-----
        //      res.setContentType("text/html");
        //      PrintWriter out=res.getWriter();
        //      out.println("<h1>hello</h1>");

        ///-----text/plain-----

        //      res.setContentType("text/plain");
        //      PrintWriter out = res.getWriter();
        //      out.println("Hello User");
        //      out.println("Welcome to Servlet");

        ///-----application/json-----

        //      res.setContentType("application/json");
        //      PrintWriter out = res.getWriter();
        //      out.print("{
//          + "\"id\":1,"
//          + "\"name\":\"Shahin\","
//          + "\"course\":\"Java\""

```



```
//          + "}");

    ////-----application/xml-----
    res.setContentType("application/xml");

    PrintWriter out = res.getWriter();

    String xml =
        "<user>" +
        "<id>1</id>" +
        "<name>Shahin</name>" +
        "</user>";

    out.print(xml);
}
}
```

Supporting Interface

Class 1:

```
package supportingInterface;

import jakarta.servlet.ServletConfig;
import jakarta.servlet.ServletContext;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;

public class SupportingInterface extends HttpServlet {
    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse
resp) throws ServletException, IOException {
        //      super.doGet(req, resp);
        ServletConfig config=super.getServletConfig();
        String username=config.getInitParameter("username");
        String password=config.getInitParameter("password");
        System.out.println(username+" - "+password);

        //
        -----

        ServletContext context=config.getServletContext();
        String url= context.getInitParameter("url");
    }
}
```

```

        System.out.println(url);
    }
}

```

Class 2:

```

package supportingInterface;

import jakarta.servlet.ServletConfig;
import jakarta.servlet.ServletContext;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

import java.io.IOException;

public class SupportingInterface2 extends HttpServlet {
    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse
resp) throws ServletException, IOException {
        //      super.doGet(req, resp);
        ServletConfig config=super.getServletConfig();
        String username=config.getInitParameter("username");
        String password=config.getInitParameter("password");
        System.out.println(username+" - "+ password);

        //
        -----
        //      ServletContext context=config.getServletContext();
        //      String url= context.getInitParameter("url");
        //      System.out.println(url);
        //      -----or-----
        ServletContext context=super.getServletContext();
        String url= context.getInitParameter("url");
        System.out.println(url);
    }
}

```

Xml.fail

```

<servlet>
  <servlet-name>A</servlet-name>

```

```

<servlet-class>supportingInterface.SupportingInterface</servlet-class>
  <init-param>
    <param-name>username</param-name>
    <param-value>root</param-value>
  </init-param>
  <init-param>
    <param-name>password</param-name>
    <param-value>Shahin@1205</param-value>
  </init-param>
</servlet>

<servlet-mapping>
  <servlet-name>A</servlet-name>
  <url-pattern>/support</url-pattern>
</servlet-mapping>

<servlet>
  <servlet-name>B</servlet-name>

<servlet-class>supportingInterface.SupportingInterface2</servlet-class>
</servlet>

<servlet-mapping>
  <servlet-name>B</servlet-name>
  <url-pattern>/support1</url-pattern>
</servlet-mapping>

<context-param>
  <param-name>url</param-name>
  <param-value>jdbc:mysql</param-value>
</context-param>

```

CRUD operation

```

package EnpoinConnection;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

```

```
import java.io.IOException;
import java.lang.reflect.Array;
import java.util.ArrayList;
import java.util.Arrays;
import java.util.List;

class UserDetails{
    private int id;
    private String name;
    private String email;
    private int price;

    public UserDetails(int id, String name, String email, int price) {
        this.id = id;
        this.name = name;
        this.email = email;
        this.price = price;
    }

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getEmail() {
        return email;
    }

    public void setEmail(String email) {
        this.email = email;
    }

    public int getPrice() {
        return price;
    }

    public void setPrice(int price) {
        this.price = price;
    }
}
```

```

@WebServlet("/user")
public class User extends HttpServlet {
    private ArrayList<UserDetails> list=new ArrayList<>(Arrays.asList(
        new UserDetails(1, "shahin", "shahin@gmail.com", 234),
        new UserDetails(2, "mahin", "mahin@gmail.com", 567),
        new UserDetails(3, "visal", "visal@gmail.com", 876),
        new UserDetails(4, "gokul", "gokul@gmail.com", 876)
    ));

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse
res) throws ServletException, IOException {
        res.setContentType("application/json");
        StringBuilder sb = new StringBuilder();
        sb.append("[");
        for (int i = 0; i < list.size(); i++) {
            UserDetails userDetails = list.get(i);
            sb.append("{").append("\"id\": " + userDetails.getId() +
",");
            sb.append("\"name\": " + userDetails.getName() + "\",");
            sb.append("\"email\": " + userDetails.getEmail() + "\",");
            sb.append("\"price\": " + userDetails.getPrice() + "\"}");
            if (i != list.size() - 1) {
                sb.append(",\n");
            }
        }
        sb.append("]");
        String jsonResponse = sb.toString();
        res.getWriter().println(jsonResponse);
    }

    @Override
    protected void doPost(HttpServletRequest req, HttpServletResponse
res) throws ServletException, IOException {
        //      super.doPost(req, resp);
        int id=Integer.parseInt(req.getParameter("id"));
        String name=req.getParameter("name");
        String email=req.getParameter("email");
        int price=Integer.parseInt(req.getParameter("price"));

        UserDetails userDetails=new UserDetails(id,name,email,price);
        list.add(userDetails);
        res.setContentType("text/plain");
        res.getWriter().println("added");
    }

    @Override

```

```

    protected void doDelete(HttpServletRequest req, HttpServletResponse
res) throws ServletException, IOException {
        int id=Integer.parseInt(req.getParameter("id"));

        res.setContentType("text/plain");

        for (int i=0;i<list.size();i++){
            if(list.get(i).getId()==id){
                list.remove(i);
                res.getWriter().println("remove success full");
                return;
            }
        }
        res.getWriter().println("user not foud");
    }

    @Override
    protected void doPut(HttpServletRequest req, HttpServletResponse
resp) throws ServletException, IOException {
        int id=Integer.parseInt(req.getParameter("id"));
        String name = req.getParameter("name");
        String email=req.getParameter("email");
        int price=Integer.parseInt(req.getParameter("price"));

        resp.setContentType("text/plain");
        for(UserDetails user: list){
            if (user.getId()==id){
                user.setName(name);
                user.setEmail(email);
                user.setPrice(price);
                resp.getWriter().println("update success full");
                return;
            }
        }
        resp.getWriter().println("user not fount");
    }
}

```

RequestDispatcher `requestDispatcher`

Class 1

```

package requestDispatcher;

```

```

import jakarta.servlet.RequestDispatcher;
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;
@WebServlet("/A")
public class ClassA extends HttpServlet {
    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse
res) throws ServletException, IOException {
//        super.doGet(req, resp);

        RequestDispatcher rd=req.getRequestDispatcher("/B");
        rd.forward(req,res);
        System.out.println("A");

        String name =req.getParameter("name");
        System.out.println(name);
    }
}

//http://localhost:8080/ServletEndpoint/A?name=sample

```

Class 2:

```

package requestDispatcher;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

import java.io.IOException;

@WebServlet("/B")
public class ClassB extends HttpServlet {
    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse
resp) throws ServletException, IOException {
//        super.doGet(req, resp);

        System.out.println("B");

        String name=req.getParameter("name");
        System.out.println(name);
    }
}

```

```
    }  
}
```

Sendredirect

Class 1:

```
package sendredirect;  
  
import jakarta.servlet.RequestDispatcher;  
import jakarta.servlet.ServletException;  
import jakarta.servlet.annotation.WebServlet;  
import jakarta.servlet.http.HttpServlet;  
import jakarta.servlet.http.HttpServletRequest;  
import jakarta.servlet.http.HttpServletResponse;  
  
import java.io.IOException;  
@WebServlet("/redirect1")  
public class redirect1 extends HttpServlet {  
    @Override  
    protected void doGet(HttpServletRequest req, HttpServletResponse  
res) throws ServletException, IOException {  
        //      super.doGet(req, resp);  
  
        res.sendRedirect("redirect2?name=sample");  
        //      System.out.println("redirect1");  
        //  
        //      String name =req.getParameter("name");  
        //      System.out.println("redirect1 - "+name);  
  
    }  
}
```

Class 2:

```
package sendredirect;  
  
import jakarta.servlet.ServletException;  
import jakarta.servlet.annotation.WebServlet;  
import jakarta.servlet.http.HttpServlet;  
import jakarta.servlet.http.HttpServletRequest;  
import jakarta.servlet.http.HttpServletResponse;  
  
import java.io.IOException;
```



```
@WebServlet("/redirect2")
public class redirect2 extends HttpServlet {
    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse
resp) throws ServletException, IOException {
        // super.doGet(req, resp);
        System.out.println("redirect2");
        String name = req.getParameter("name");
        System.out.println("redirect2 - "+name);
        // resp.getWriter().print(name);
    }
}
```